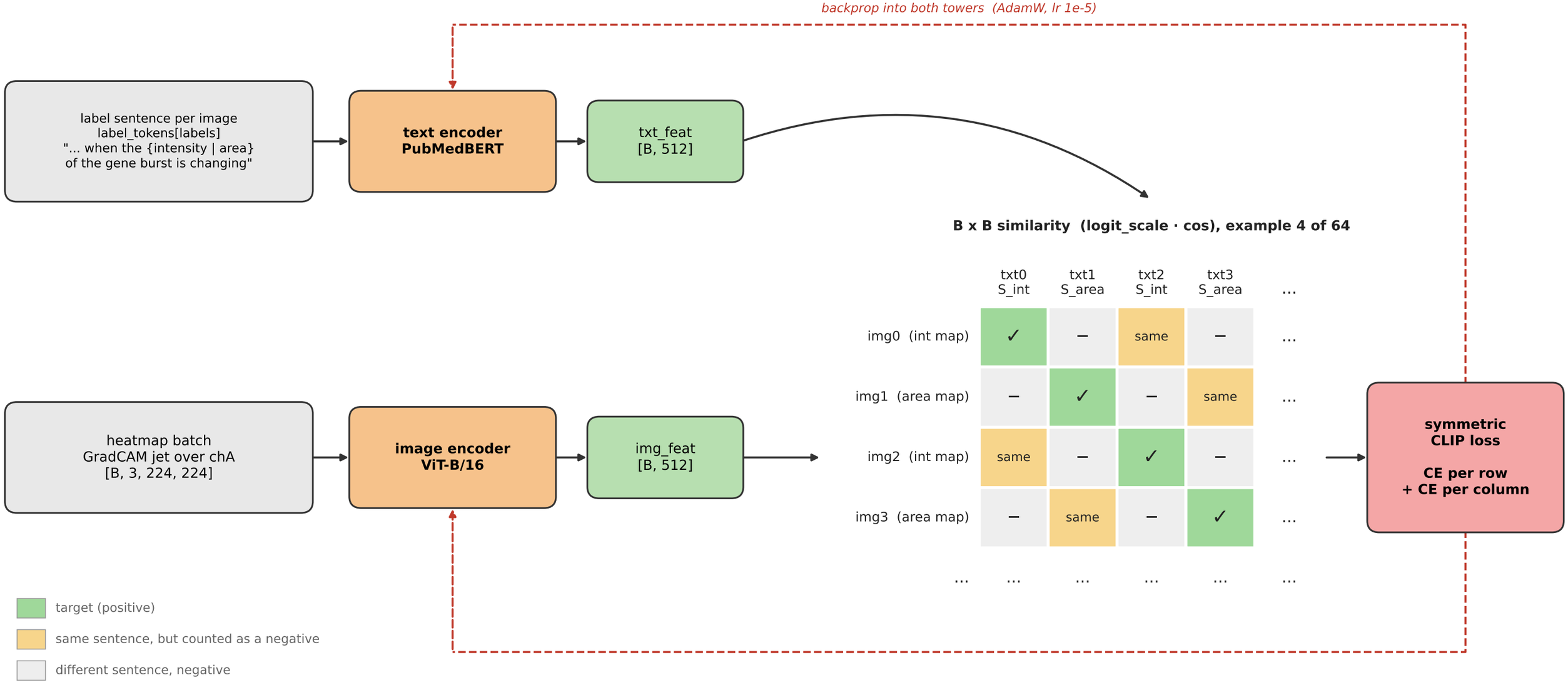
BiomedCLIP finetuning on GradCAM heatmaps (training)
backprop into both towers (AdamW, lr 1e-5)
label sentence per image
label_tokens[labels]
"... when the {intensity | area}
of the gene burst is changing"
text encoder
PubMedBERT
txt_feat
[B, 512]
heatmap batch
GradCAM jet over chA
[B, 3, 224, 224]
image encoder
ViT-B/16
img_feat
[B, 512]
B x B similarity (logit_scale · cos), example 4 of 64
txt0
S_int
txt1
S_area
txt2
S_int
txt3
S_area
img0 (int map)
img1 (area map)
img2 (int map)
img3 (area map)
same
symmetric
CLIP loss
CE per row
+ CE per column
target (positive)
same sentence, but counted as a negative
different sentence, negative